Q-CLUSTER

Platform for Enterprise Quantum Adoption

Drive from quantum exploration to real-world application; with existing expertise, infrastructure, and a focused use case. Q-Cluster brings the essential quantum development capabilities into one integrated environment, enabling organizations to learn, experiment, develop, benchmark, and execute quantum workloads without building a complex quantum technology stack from the ground up.

Q-Cluster enables organizations to establish their quantum computing capabilities around any specific business and engineering problems. Teams can build and validate quantum solutions, evaluate their performance, and benchmark them against classical approaches; all within a unified environment. Rather than investing effort in assembling and managing a fragmented quantum technology stack, teams can focus their resources on identifying, developing, and validating meaningful applications.

As quantum solutions evolve, Q-Cluster enables workloads to run seamlessly across diverse computing environments, from CPU and GPU systems to HPC infrastructure and QPUs. This provides a consistent development environment while allowing organizations to leverage the computing resources best suited to their requirements.

Beyond individual algorithms and experiments, Q-Cluster provides an application-oriented development architecture through Quantum Projects. The combination of multi-model simulation, hardware abstraction, application-level project management, and collaborative development positions Q-Cluster as a development and execution environment for building complete quantum applications; not simply a tool for running quantum circuits.

Product Description

Q-Cluster brings the complete quantum application development stack into a unified environment, combining circuit programming, multi-modal simulation, noise and error analysis, optimization, benchmarking, validation, and execution. Unlike platforms focused on individual quantum workflows, Q-Cluster provides multiple simulation approaches, including statevector, density-matrix, and tensor-network methods within the same interface. Users can select the appropriate computational method based on circuit characteristics and resource requirements. Its hardware-agnostic execution layer abstracts the underlying infrastructure, enabling the same quantum workload to be developed and evaluated across CPU, GPU, and HPC resources, while integrating with leading quantum and classical frameworks such as Qiskit, PennyLane, PyTorch, Pandas, etc. This allows organizations to maintain a consistent development workflow even as the underlying compute architecture changes. It also supports execution on QPU resources, including platforms such as IBM, IQM and many more.

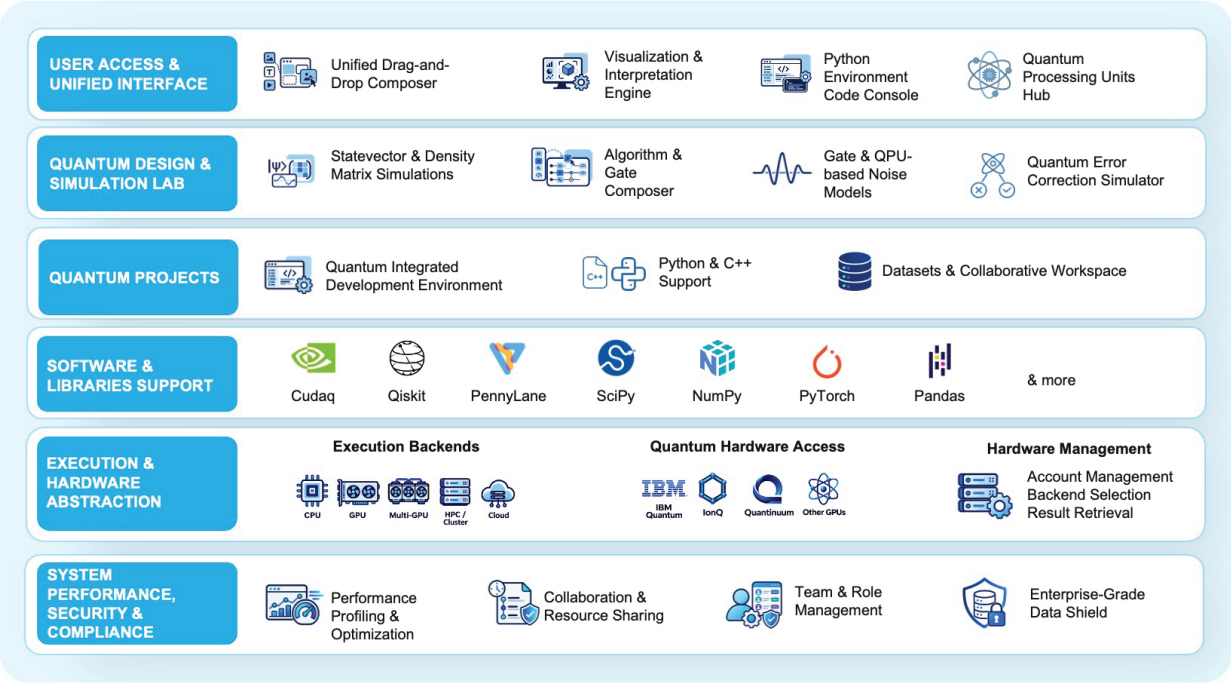


Fig 1: Q-Cluster Technology Stack

To extend this capability beyond individual development, Quantum Projects enables teams to collaborate, organize application workflows, and build reusable quantum assets. This helps transform individual experiments into a structured and growing internal quantum capability. The result is a practical pathway for organizations to explore quantum computing, validate focused use cases, and progressively build toward real-world quantum applications. Q-Cluster can be applied to emerging quantum opportunities across domains such as Portfolio Optimization, Hyperspectral Imaging, Ground State Energy Estimation, and many more.

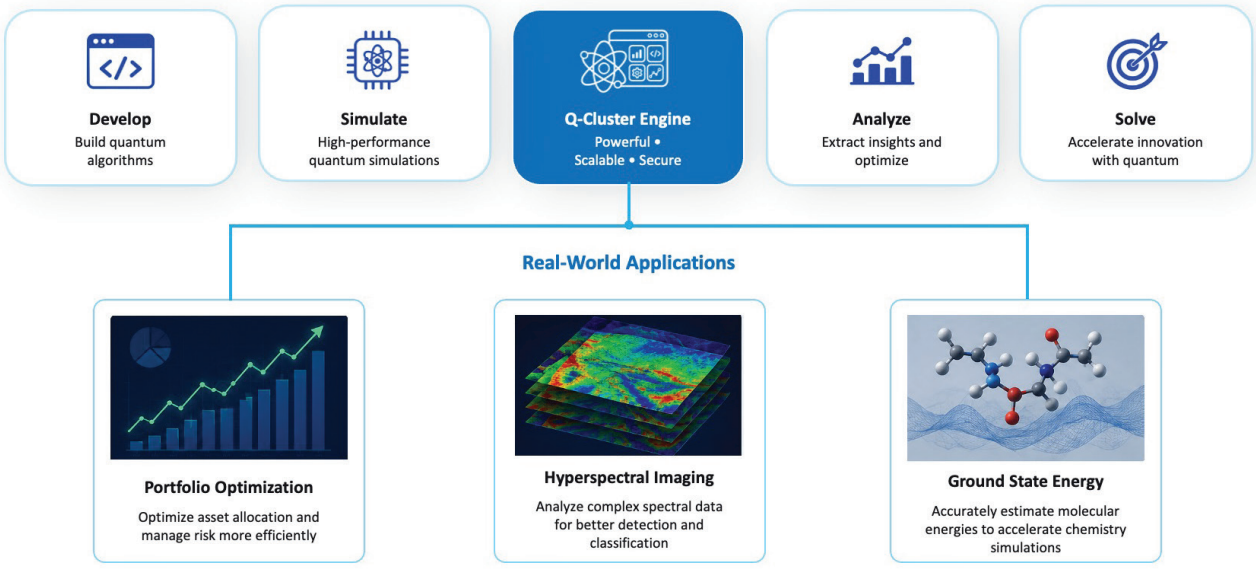


Fig 2: Workflow of Quantum Projects

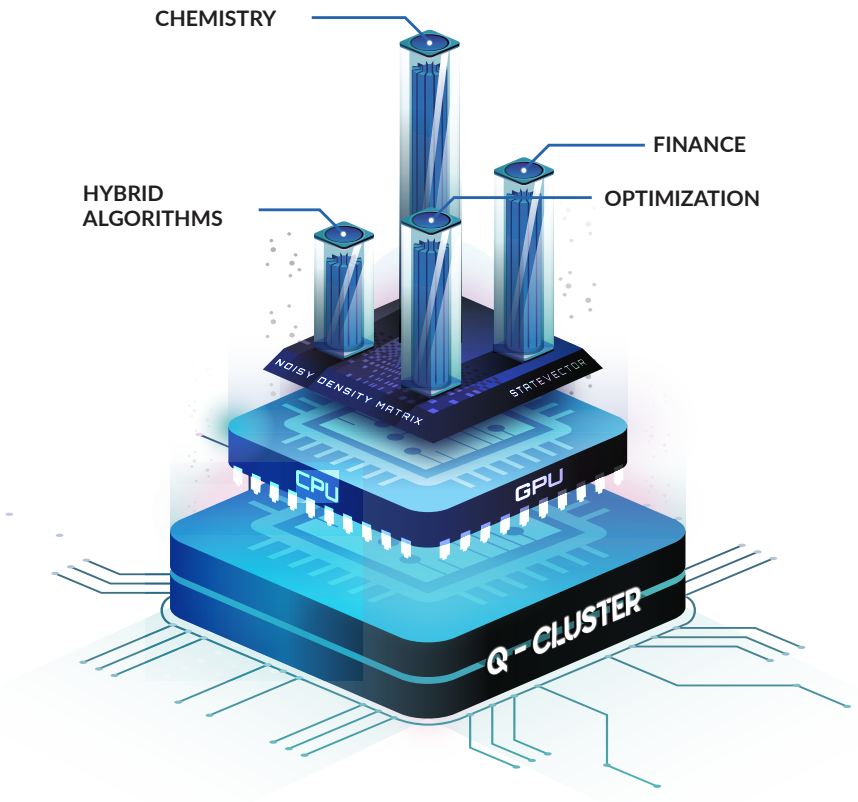
Technical Specifications

S.no	Specification	Description
1	Scalable Computing Infrastructure	CPU, GPU, HPC (Server based - T4/V100/A100/H100)
2	Simulation Types	Statevector, Noisy Density Matrix, TensorNet
3	Supported Quantum Algorithms	Complete spectrum: Deutsch's, Grover's, Shor's, QPE, HHL, and more
4	Noise Models	Gate errors, decoherence, readout noise, several in-built QPU-approximated models)
5	QPU Hub	Dedicated circuit composer for automatic transpilation and execution of quantum circuits on real quantum hardwares
6	Qubits Simulatable	**Statevector:** 28 - Nvidia RTX 6000 Pro based Server 32 - Nvidia 1 x H100 based Server 36 - Nvidia 4 x H100 based Server **Tensor Networks:** 100-120 (depending on the circuit configuration) - Nvidia RTX 5070 based Server
7	Educational Mode	Basics of quantum computing, tutorials of quantum algorithms along with intuitive interpretation of quantum algorithms, guided modules for approximation of QPU behavior, understanding bloch sphere
8	Quantum Error Correction	Simulation and analysis of logical qubit encoding/decoding schemes, and noise-aware circuit performance assessment

9	Multi-user Environment	Enables simultaneous access and collaborative use by multiple users
10	Quantum Projects (Quantum-dedicated Integration Development Environment)	VSCode-inspired workspace for collaborative development, validation, and management of quantum algorithms and domain-specific applications within a unified environment
11	Quantum Algorithm Application	Quantum Finance, Medical and Hyperspectral imaging, Quantum Chemistry and more

Key Features

- **Project-Based Development:** Create and organize multiple quantum projects, each containing source code, quantum circuits, datasets, notebooks, documentation, experiment results, and configuration files.
- **Collaborative Workspaces:** Work with Multiple users and teams on interconnected projects; enabling seamless collaboration, knowledge sharing, and coordinated development across research groups and organizations.
- **Middleware for Quantum Development:** Serves as the central development layer between quantum SDKs, simulators, GPUs, and quantum hardware, allowing users to develop, validate, optimize, and execute quantum algorithms through a unified workflow.
- **Broad SDK & Library Support:** Supports both quantum and classical-based packages such as CUDA-Q, Qiskit, PennyLane, Cirq, OpenQASM, PyTorch, NumPy, SciPy, and other Python libraries. Additional packages can be installed within individual projects to support specialized applications.
- **Application-Oriented Development:** Enables development of quantum and hybrid solutions across domains such as financial portfolio optimization, hyperspectral image classification, quantum chemistry simulation, and medical image classification.
- **Algorithm Development & Validation:** Develop, benchmark, and validate quantum algorithms using integrated simulation, performance analysis, and hardware execution capabilities.
- **Scalable Execution:** Execute workloads on CPUs, GPUs, multi-GPU systems, and supported quantum hardware through a common interface while maintaining experiment history and reproducibility.



Turn quantum adoption into action. Develop around a focused use case. Validate before scaling. Build on the infrastructure you already have. Turn quantum potential into practical value.